LECTURE ACTIVITY 13: MURDEROUS NURSE? – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

This activity is heavily adapted from one created by the STAT 1110 Course Coordination Team, which in turn is modified from "Introduction to Statistical Investigations," by Tintle et. al.

Learning Objectives:

By the end of this activity, you should be able to

1) Organize counts into a two-way table, when data are available on two categorical variables for the same set of observational units.
2) Calculate conditional proportion of successes, for different categories of the explanatory variable
3) Use conditional proportions to decide whether there is preliminary evidence of an association between the explanatory and response variables.
4) Create a segmented bar chart and mosaic plots to display data available on two categorical variables for the same set of observational units.
5) Calculate and interpret the difference in conditional proportions.
6) Calculate and interpret the relative risk.
7) Calculate and interpret the odds ratio.

Key Terms:

- Two-way table
- Conditional proportion
- Segmented bar chart
- Mosaic plot
- Relative risk
- Odds ratio

For several years in the 1990s, Kristen Gilbert worked as a nurse in the intensive care unit (ICU) of the Veterans Administration Hospital in Northampton, Massachusetts. Over the course of her time there, other nurses came to suspect that she was killing patients by injecting them with the heart stimulant epinephrine. Gilbert was eventually arrested and charged with these murders. Part of the evidence presented against Gilbert at her murder trial was a statistical analysis of 1,641 eight-hour shifts during the time Gilbert worked in the ICU. For each of these shifts, researchers recorded two variables: whether Gilbert worked on the shift and whether at least one patient died during the shift.

Question 1

Identify an observational unit in this study. (**Hint**: An observational unit is not a person. Think about what the sample size is.)

Solution: One observational unit is one shift.

Question 2

Classify each variable as categorical or quantitative and determine which is the explanatory variable and which is the response variable.

(a) **Whether or not Gilbert worked on the shift:**

Select One: √ **Categorical** ◯ Quantitative

Select One: √ **Explanatory** ◯ Response

(b) **Whether or not at least one patient died during the shift:**

Select One: √ **Categorical** ◯ Quantitative

Select One: ◯ Explanatory √ **Response**

Question 3

Is this an observational study, or an experiment? Justify your answer.

Solution: This is an observational study because the researchers did not assign Gilbert to work or not work on particular shifts; rather, they simply observed what happened during existing shifts.

So, what did the researchers find when they analyzed the 1,641 eight-hour shifts? They found that Gilbert worked during 257 of the shifts, and she did not work during the remaining 1384. They also found that at least one patient died during 74 of the shifts, and no death occurred during the other 1,567 shifts. Among the 74 shifts during which at least one patient died, 40 were shifts on which Gilbert worked, and 34 were shifts on which Gilbert did not work.

Concept Overview

When analyzing data on two categorical variables, a useful first step is to organize the data into a **two-way table** of counts. In this case both variables have only 2 categories, so we produce a 2×2 table.

Question 4

Fill in the missing counts in the following two-way table. **Hint:** Start by entering 40 and 34 into the appropriate cells. Then make sure that both rows and both columns add up to the indicated totals.

	Gilbert working on shift	Gilbert not working on shift	Total
At Least One Patient Died During the Shift	**40**	**34**	**74**
No Patients Died During the Shift	**217**	**1350**	**1567**
Total	**257**	**1384**	**1641**

Question 5

Suggest a better way to compare the outcomes of the shifts on which Gilbert worked and the shifts on which she did not work than simply comparing 40 to 34. Explain why your answer is more informative.

Solution: Gilbert worked far fewer shifts than the non-Gilbert nurses, so comparing counts alone is misleading. A better comparison is to compare the proportions of shifts with deaths when Gilbert is working and when Gilbert is not working rather than the counts 40 and 34.

Concept Overview

The next step in analyzing categorical data in a two-way table is to calculate **conditional proportions**. A conditional proportion simply means that you consider the counts in only one category of the explanatory variable at a time.

Question 6

(a) Calculate the conditional proportion of Gilbert shifts on which at least one patient died. (**Hint**: Put the total number of Gilbert shifts in the denominator of your fraction.)

Solution: $\frac{40}{257} \approx \mathbf{0.156}$. In other words, around 15.6% of Gilbert's shifts had at least one patient death occur in them.

(b) Calculate the conditional proportion of non-Gilbert shifts on which at least one patient died.

Solution: $\frac{34}{1384} \approx \mathbf{0.025}$. In other words, around 2.5% of non-Gilbert shifts had at least one patient death occur in them.

Visual Explorations

In many cases when comparing two variables, a good set of visualizations is enough to be able to start to see patterns. That is, even without sophisticated statistical tools, we can start to examine whether or not there is an association between whether or not Gilbert was on shift and whether or not at least one patient died.

Let's explore a few of these visualizations now.

Segmented Bar Graph

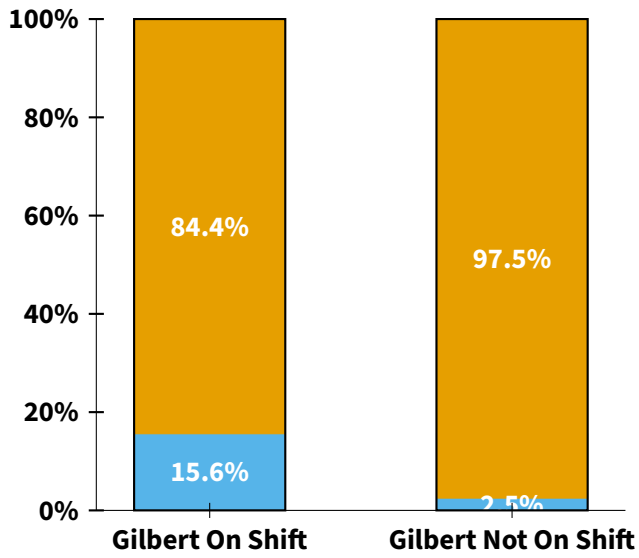
Concept Overview

A **segmented bar graph** is an appropriate display for graphing data in a two-way table. Such a graph contains rectangles of total height 100% for each category of the explanatory variable. Then segments divide up each rectangle according to the conditional proportions of the response variable categories. With this type of graph, you can display the association between the two categorical variables by showing the conditional proportions of success and failure across the explanatory variable groups.

- If the variables have absolutely no association, then the breakdown of segments is identical for across all of the bars.
- If the variables have absolutely no association, the conditional proportions are identical for all explanatory variable groups.

Question 7

Fill in the following segmented bar graph to display the data from this study. First draw two rectangles, one for Gilbert shifts and one for non-Gilbert shifts, each with total height 100%. Then draw a line in each rectangle corresponding to the conditional proportion of shifts in which a patient died, as you calculated in Question (4) above.



Mosaic Plots

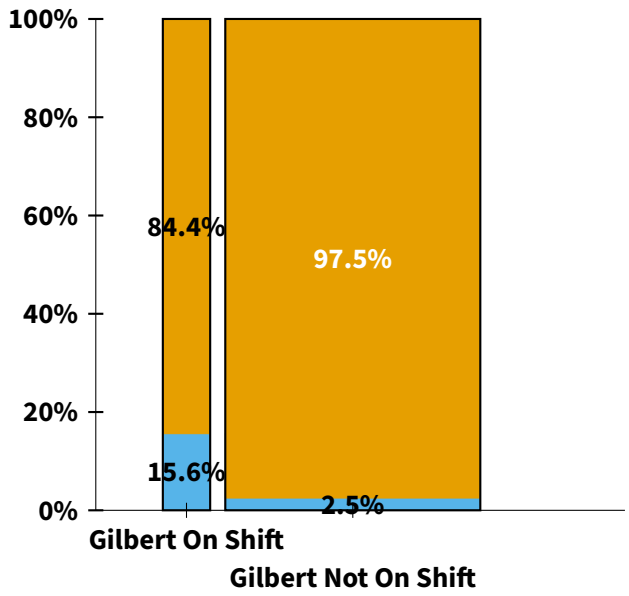
Concept Overview

Another visual is a **mosaic plot**, which allows the widths of the bars to represent the size of each group in the sample. A mosaic plot also makes the widths of the rectangles proportional to the sizes of the categories.

Question 8

Fill in a mosaic plot to display the data from this study:

- Start by drawing two rectangles, one for Gilbert shifts and one for non-Gilbert shifts, but make the rectangles have different widths.
  - Specifically, make whichever category (Gilbert on Shift, or Gilbert Not on Shift) have a smaller width.
- Then draw a line in each rectangle corresponding to the conditional proportion of shifts in which a patient died, as you calculated in Question (4) above.



Question 9

Do the conditional proportions, the segmented bar graph, and the mosaic plots appear to provide evidence that at least one patient dying was more likely to happen on a Gilbert shift than on a non-Gilbert shift? In other words, does there appear to be a tendency for Gilbert shifts to have at least one death more often than non-Gilbert shifts? Explain.

Solution: Yes. The conditional proportion of deaths is much higher during Gilbert shifts, 15.6%, than during non-Gilbert shifts, 2.5%. This suggests an association between Gilbert working and patient deaths.

Question 10

Without doing any further analysis, do you consider the difference between the conditional proportions to be striking, worth reporting to a jury?

Solution: The difference appears large and would likely be worth reporting because the death rate is much higher on Gilbert shifts. But it is important to remember that we do not know yet if these results could have happened by chance alone.

Numerical Explorations

A next step in this analysis is to produce a single number (statistic) to summarize the data. We will cover 3 different statistics in this part of the activity: difference in sample proportions, relative risk, and odds-ratio.

Statistic 1: Difference in Proportions

So far in this activity we have obtained two summary measures:

- The conditional proportion of Gilbert shifts on which at least one patient died: 40/257 = 0.156
- The conditional proportion of non-Gilbert shifts on which at least one patient died: 534/1,384 = 0.025

If we wanted to change these 2 numbers into a single statistic, we could just compute the **difference** in these proportions.

Question 11

Calculate the difference in conditional proportions of death between Gilbert and non-Gilbert shifts. Does the value of this difference strike you as noteworthy?

Solution: Difference in proportions: $0.156 - 0.025 = \mathbf{0.131}$. The difference is approximately 0.131, or 13.1 percentage points, which seems noteworthy.

Question 12

Write a sentence interpreting the value you obtained in Question (11) above.

Solution: The proportion of shifts with at least one death was about **13.1 percentage points higher** when Gilbert was working than when she was not working.

Statistic 2: Relative Risk

Another way that we can summarize the two proportions into a single statistic is to divide one proportion by the other.

Concept Overview

Relative risk is the ratio of two conditional proportions. It indicates how many times greater the risk of an outcome is for one group compared to another.

Question 13

Now calculate the relative risk. **Tip:** put the larger value in the numerator.

Solution: Relative risk: $\frac{0.156}{0.025} = \mathbf{6.24}$

Question 14

Write a sentence interpreting the value you obtained in Question (13) above.

Solution: The risk of at least one patient death was about 6.24 times higher during Gilbert shifts than during non-Gilbert shifts.

One drawback to using relative risk is that it can convey different impressions of how strong the relationship is depending on how the calculation is set up. That is, the value of the relative risk depends on which variable is explanatory, which is response, and which category is called "success." This can be confusing.

Statistic 3: Odds Ratio

In certain observational studies, samples of subjects are taken based on their status according to the *response* variable. Such studies are called **case-control** studies. Because the sample sizes in a case control study are dependent on the response variable, examining differences in proportions and relative risk ratios doesn't make a whole lot of sense (at least, in terms of trying to get a real-world risk associated with Gilbert being on shift). Instead, we use a new type of statistic: the odds ratio.

Let's build up to the notion of an odds ratio. First recall that the **proportion** of successes (in a group) is a ratio between the number of successes and the total number of observations (in the group):

$$\text{proportion of success} = \frac{(\text{number of successes})}{(\text{number of successes}) + (\text{number of failures})} = \hat{p}$$

The **odds** of success is a ratio between the number of successes *and the number of failures*:

$$\text{odds of success} = \frac{(\text{number of successes})}{\text{number of failures}} = \frac{\hat{p}}{1-\hat{p}}$$

Concept Overview

The **odds ratio** is the ratio of odds:

$$\text{odds ratio} = \frac{\text{odds for group 1}}{\text{odds for group 2}}$$

- The odds ratio indicates how many times greater the odds of success are for one group compared to another
- A major advantage of the odds ratio is that any 2×2 table has only one possible value of the odds ratio. That is, the odds ratio does not depend on the explanatory/response distinction, or the success/failure distinction (assuming you follow the convention that the calculation is set up so the odds ratio is at least 1).
- The disadvantage of the odds ratio is that it's harder to interpret ("higher odds" instead of the more natural "more likely").

Question 15

Calculate and interpret the odds ratio comparing the odds of death for the Gilbert shifts to the odds of death for the non-Gilbert shifts. Interpret this value.

Solution:

$$\text{odds ratio} = \frac{\text{Odds of death during Gilbert shifts}}{\text{Odds of death during non-Gilbert shifts}} = \frac{\frac{40}{217}}{\frac{34}{1350}} = \mathbf{7.32}$$

The odds of at least a patient death were about 7.32 times greater during Gilbert shifts than during non-Gilbert shifts.

Question 16

Calculate and interpret the odds ratio comparing the odds of staying alive for the Gilbert shifts to the odds of staying alive for the non-Gilbert shifts. Interpret this value.

Solution:

$$\text{odds ratio} = \frac{\text{Odds of staying alive during Gilbert shifts}}{\text{Odds of staying alive during non-Gilbert shifts}} = \frac{\frac{217}{40}}{\frac{1350}{34}} = 0.13663 = \frac{1}{7.32}$$

Swapping the notion of "success" has the effect of flipping the odds ratio.

Question 17

Even if the difference between these conditional proportions of death is found to be highly unlikely to have happened by chance, is it reasonable to conclude that Kristen Gilbert was responsible for (i.e., caused) the increased chance of death on her shifts? Explain your answer. (**Hint**: Remember your answer to Question (3) about what kind of study this is.)

Solution: No. Even if the difference is statistically significant, we cannot conclude that Gilbert caused the increased death rate because this was an observational study, so there can be confounding variables affecting the results of the study. Association does not prove causation.

Question 18

Suggest an alternative explanation that the defense attorneys might use to explain the large difference in death proportions between the two groups.

Solution: A possible alternative explanation is that Gilbert could have been working with more critically ill patients or may she worked shifts during times when deaths were more likely (night).

Question 19

(**Optional**) Was it enough? That is, navigate to Google (or your favorite search engine) and see what the "final" outcome of the trial was.

Solution: From Wikipedia: "Kristen Heather Gilbert (née Strickland; born November 13, 1967) is an American serial killer and former nurse who was convicted of four murders and two attempted murders of patients admitted to the Veterans Affairs Medical Center (VAMC) in Northampton, Massachusetts."